



EPD-TRH-INT

Gebruikershandleiding/Manual

ATAL B.V.
Ampèrestraat 35-37
NL-1446 TR PURMEREND

Postbus 783
NL-1440 AT PURMEREND

T (+31) 0299 630 610
F (+31) 0299 630 611

E info@atal.nl
I www.atal.nl

Table of contents

Introduction.....	3
General safety rules	3
Device description and important notices.....	4
Getting started	6
What is needed for operation.....	6
Mounting the device.....	6
Device settings	7
Checking functions	8
Device setup	9
Setup using web interface.....	9
Setup using TSensor software.....	19
Factory defaults	20
Communication protocols	22
Website	22
SMTP – sending e-mails	22
SNMP	22
Modbus TCP	25
SOAP	26
Syslog	28
SNTP	28
Troubleshooting	29
I forgot the device IP address	29
I cannot connect to the device.....	29
I forgot the password for setup	29
Factory defaults	29
Display is off.....	30
Device error states	31
Technical specifications.....	32
Dimensions.....	32
General parameters	34
Measured values	35
Temperature measurement.....	36
Relative humidity measurement.....	36
Atmospheric pressure measurement	37
Computed quantities	37
Operating terms.....	39
End of operation	40
Technical support and service.....	40
Preventive maintenance	40
Optional accessories.....	41

Introduction

This chapter provides basic information about device. Before starting please read this manual carefully.

The Web Sensor devices are designed for measurement of temperature, relative humidity and atmospheric pressure in air without aggressive substances. Measured quantities are according device type. Devices with relative humidity measurement can show one of computed quantities: dew point temperature, absolute humidity, specific humidity, mixing ratio and specific enthalpy. Temperature units are °C or °F. Pressure units are hPa, PSI, inHg, mBar, oz/in², mmHg, inH₂O and kPa. Communication interface of Web Sensors is Ethernet. Device types:

Type	Temperature	Relative humidity	Pressure	Computed quantity
EPD-T-INT	✓			
EPD-TRH-INT	✓	✓		✓
EPD-TRHP-INT	✓	✓	✓	✓
EPD-P-INT			✓	
EPD-TRH(P)	✓	✓		✓
EPD-T	✓			
EPD-TRHP	✓	✓	✓	✓

Web Sensor EPD-xxx-xxx-P devices are designed for measurement in compressed air up to 25 bars. The removable probe is integral part of the device. It is not possible to interchange probe between devices. Probe has connector with protection IP67.

Models marked EPD-xxx-Z are custom-specified devices. Description of these devices is not included in this manual.

General safety rules

The following summary is used to reduce the risk of injury or damage the device. To prevent injury, please follow instructions.



The device can be serviced only by a qualified person. The device contains no serviceable parts inside.

Don't use the device, if it doesn't work correctly. If you think, that the device is not working correctly, let check it by qualified service person.

It's forbidden to use the device without the cover. Inside the device can be a dangerous voltage and may be risk of electric shock.

Use only the appropriate power supply adapter according to manufacturer specifications and approved according to relevant standards. Make sure, that the adapter does not have damaged cables or covers.

Connect the device only to network parts approved according to relevant standards.

Connect and disconnect the device properly. Don't connect or disconnect Ethernet cable or probes, if the device is powered.

The device may be installed only in prescribed areas. Never expose the device to higher or lower temperatures than is allowed. The device has not improved resistance to moisture. Protect it from dripping or splashing water and do not use at areas with condensation.

Devices are not designed for locations with chemically aggressive environment. Temperature and humidity sensors must not be exposed to direct contact with water or other liquids. It is not allowed to remove the sensor cover to avoid any mechanical damage of the sensors.

Before you removing probe of the Web Sensor EPD-xxx-xxx-P, make sure that pressure in a pressure chamber and ambient pressure are equal.

Don't use device in explosive environments.

Don't stress the device mechanically.

Device description and important notices

This chapter contains information about basic features. Also there are important notices concerning to functional safety.

Measured values can be displayed on LCD display or can be read using an Ethernet connection. The following formats are supported:

- Web pages with user changeable look and XML files
- Modbus TCP protocol
- SNMPv1 protocol
- SOAP protocol

The device can also be used to check measured values and if the limit is exceeded, device sends warning messages. Possible ways to sending warning messages:

- Sending e-mails up to 3 e-mail addresses
- Sending SNMP traps up to 3 configurable IP addresses
- Displaying the alarm status on web page
- Sending messages to Syslog server

The device setup can be made by the TSensor software or web interface. TSensor software can be free downloaded from the manufacturer's website. Also you will find there latest firmware for your device. Do not upload to your device firmware which is not designed for it. Unsupported firmware can damage your device.

Device does not support powering over Ethernet cable (PoE). PoE splitter must be used. Compatible PoE splitter can be purchased as optional accessories. Splitter must have 12V output with approximately 1W.



Reliability of warning messages delivering (e-mail, trap, syslog), depends on actual availability of necessary network services. The device should not be used for critical applications, where malfunction could cause to injury or loss of human life. For highly reliable systems, redundancy is essential. For more information please see standard IEC 61508.



Never connect the device directly to the Internet. If it is necessary connect the device to the Internet, properly configured firewall must be used. Firewall can be partially replaced by the NAT.

Getting started

Here you can find information necessary to put newly purchased equipment to operation. This procedure is only informative.

What is needed for operation

To install the unit you need to the following equipment. Before installation check if it's available.

- Web Sensor Tx5xx device
- power supply adapter 9 - 30V/200mA
- RJ45 LAN connection with appropriate cable
- free IP address in your network
- for Web Sensor EPD-T is required temperature probe Pt1000/3850ppm, connected by shielded cable with maximum length 10m

Mounting the device

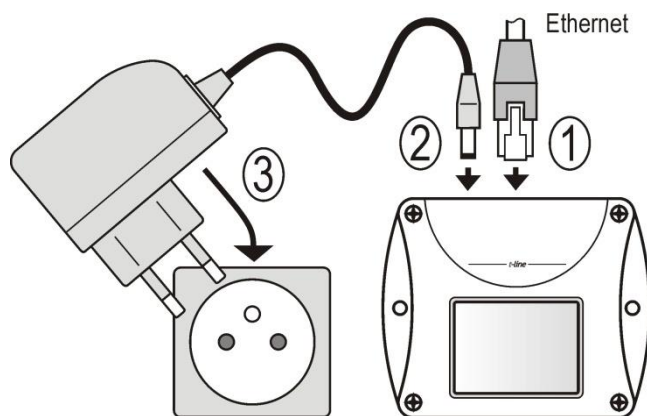
- check if the equipment from previous chapter are available
- install the latest version of TSensor software. This software is used to all device settings.
- TSensor software can be free downloaded from the manufacturer's website. Software can be also supplied on CD. Device configuration can be made using web interface. For web configuration is not TSensor software necessary.
- contact your network administrator to obtain following information for the connection to the network:

IP address:	_____
Default gateway:	_____
DNS server IP:	_____
Subnet mask:	_____

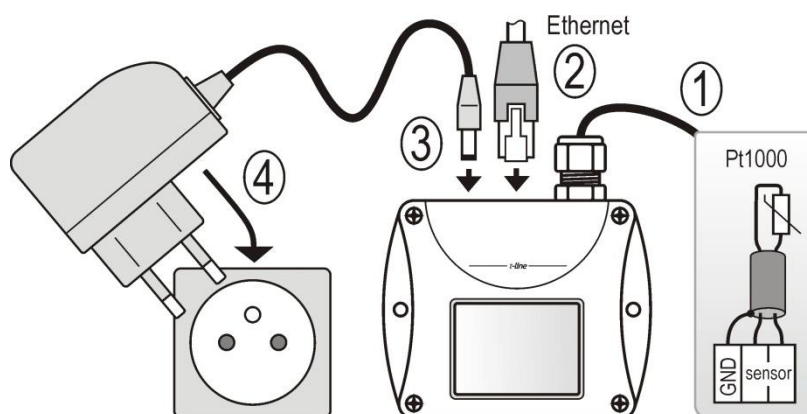
- check if there is no IP address conflict when you connect the device into network for the first time. The device has from factory set the IP address to **192.168.1.213**. This address must be changed according to information from the previous point. When you installing several new devices, connect them to the network one after another.
- connect probe Pt1000 to Web Sensor EPD-T

- connect the Ethernet connector
- connect the power adapter 9 - 30V (e.g. 12V/200mA)
- LEDs on LAN connector should blink after connecting the power and LCD start showing measured values.

Web Sensor EPD-T-INT', EPD-TRH-INT', EPD-TRHP-INT', EPD-P-INT', EPD-TRH(P), EPD-TRHP:

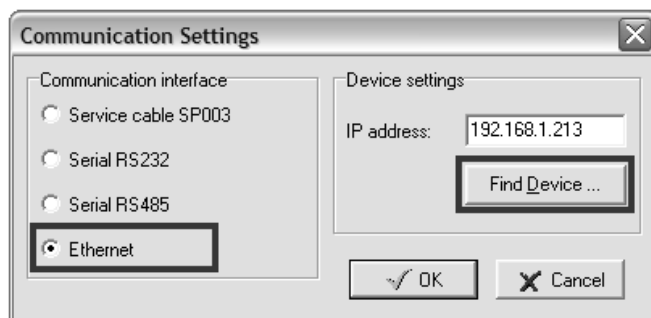


Web Sensor EPD-T:

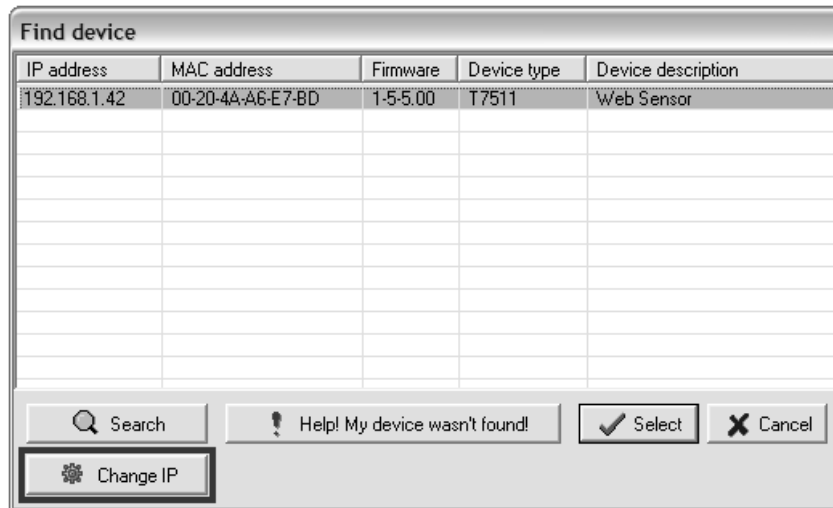


Device settings

- run configuration software TSensor on your PC
- switch to an **Ethernet** communication interface
- press button **Find device...**



- the window shows all available devices on your network



- click to **Change IP** to set new address according to network administrator instructions. If your device is not listed, then click **Help! My device wasn't found!** Then follow the instructions. MAC address is on product label. The device is factory set to IP **192.168.1.213**.



- gateway may not be entered if you want to use the device only in local network. If you set the same IP address which is already used, the device will not work correctly and there will be collisions on the network. If the device detects a collision of IP address then reboot is performed automatically.
- after changing IP address device is restarted and new IP address is assigned. Restart of the device takes about 10 seconds.
- connect to device using TSensor software and check the measured values. Set the other parameters (alarm limits, SMTP server, etc.). Settings are saved after click on button **Save changes**.

Checking functions

The last step is to check measured values on the device website. In the address bar of the web browser, enter the device IP address. If the default IP address was not changed, then insert **http://192.168.1.213**.

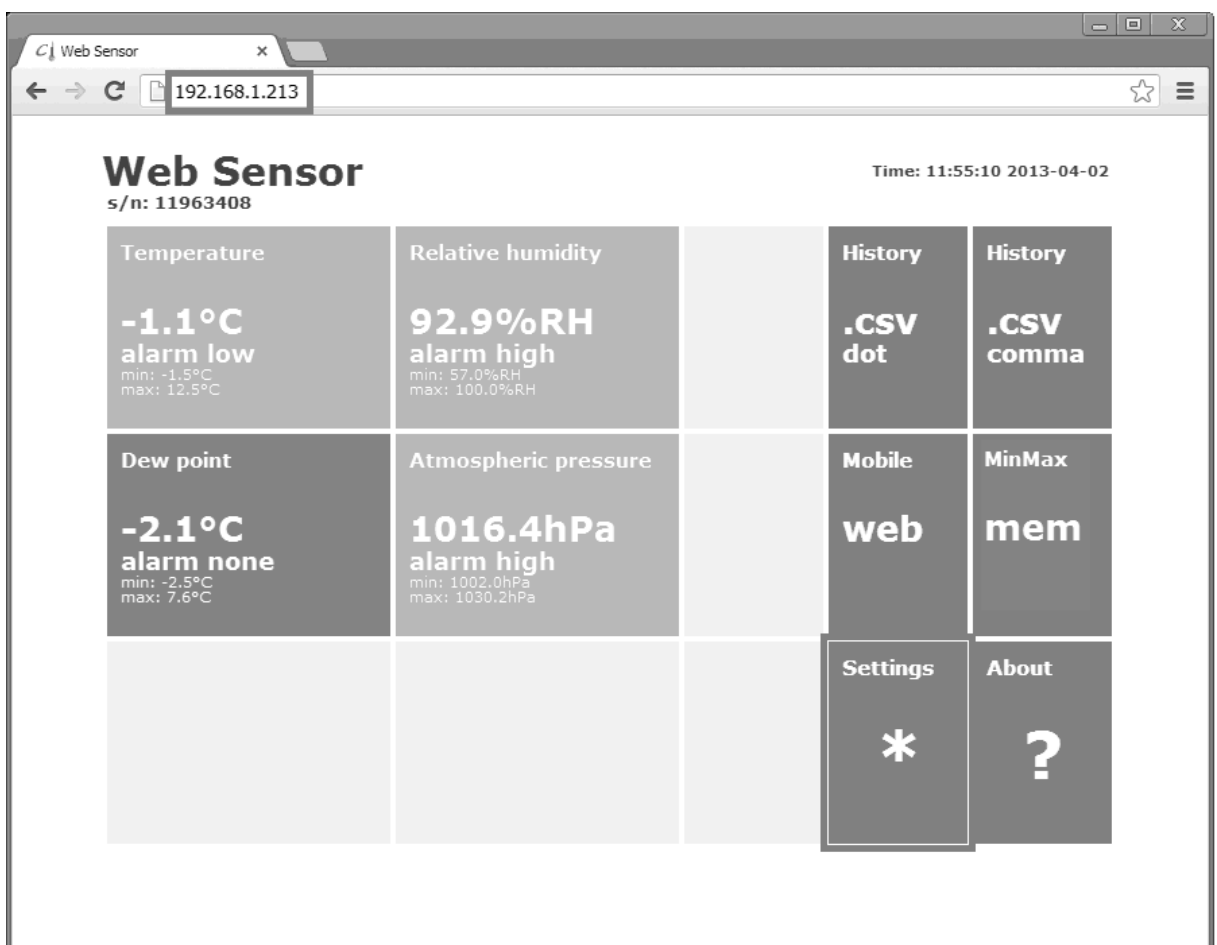
Displayed web page lists actual measured values. If the web pages are disabled, you can see text **Access denied**. In the case of measurement error a text **Error** is shown instead value.

Device setup

This chapter describes basic device configuration. There is a description of settings using web interface.

Setup using web interface

Device can be setup using web interface or TSensor software. Web interface can be managed by the web browser. Main page will be shown when you insert device address into address bar of your web browser. There you find actual measured values. Page with history graphs is shown when you click to tile with actual values. Access to device setup is possible via tile **Settings**.



General

Device name can be changed using item **Device name**. Measured values are stored into memory according **History storage interval** field. After changing of this interval all history values will be cleared. Changes must be confirmed by **Apply settings** button.

Back
Exit to main menu

Settings

General

General
General device settings

Network
Basic settings of the network interface

Alarm limits
Configuration of the alarm limits

Measuring
Units, display settings, etc.

SOAP protocol
Setup SOAP protocol for database system

Email
Alarm emails configuration

Protocols
Syslog and ModbusTCP protocol settings

SNMP
SNMPv1 protocol and SNMP Traps

Time
Synchronization with NTP server

WWW and Security
Web server and Security configuration

MinMax memory
Timestamps, clear memory

Maintenance
Factory defaults, info, etc.

General device settings. Device name can be changed to user specified name. History graphs and tables will be cleared after changing history storage interval.

Device name

Web Sensor

History storage interval

1 Min

Apply settings

Cancel changes

Network

Network parameters can be obtained automatically from DHCP server using option **Obtain an IP address automatically**. Static IP address is configurable via field **IP address**. It is not necessary to setup **Default gateway** while you use the device inside one subnet only. **DNS server IP** is required to set for proper function of DNS. Option **Standard subnet mask** sets network mask automatically according to A, B or C network class. **Subnet mask** field must be set manually, when network with nonstandard range is used. **Periodic restart interval** enables to restart device after selected time since device start.

Back
Exit to main menu

Settings

Network

General
General device settings

Network
Basic settings of the network interface

Alarm limits
Configuration of the alarm limits

Measuring
Units, display settings, etc.

SOAP protocol
Setup SOAP protocol for database system

Email
Alarm emails configuration

Protocols
Syslog and ModbusTCP protocol settings

SNMP
SNMPv1 protocol and SNMP Traps

Time
Synchronization with NTP server

WWW and Security
Web server and Security configuration

MinMax memory
Timestamps, clear memory

Maintenance
Factory defaults, info, etc.

Basic settings of the network interface. IP address can be set as either static or obtained automatically by the DHCP server. After saving the setting, device will be rebooted automatically. Before changing IP address, please contact your network administrator.

Obtain an IP address automatically☐

IP address

Default gateway

DNS server IP

Standard subnet mask☐

Subnet mask

Periodic restart interval

Apply settings

Cancel changes



11

Alarm limits

For each measurement channel is possible to set upper and lower limits, time-delay for alarm activation and hysteresis for alarm clearing.

Back
Exit to main menu

Settings

Alarm limits

General
General device settings

Network
Basic settings of the network interface

Alarm limits
Configuration of the alarm limits

Measuring
Units, display settings, etc.

SOAP protocol
Setup SOAP protocol for database system

Email
Alarm emails configuration

Protocols
Syslog and ModbusTCP protocol settings

SNMP
SNMPv1 protocol and SNMP Traps

Time
Synchronization with NTP server

WWW and Security
Web server and Security configuration

MinMax memory
Timestamps, clear memory

Maintenance
Factory defaults, info, etc.

Configuration of the alarm limits. The safe range is between high and low limit. Alarm condition occurs while measured value is out of the safe range for selected time delay. Alarm is cleared if measured value returns to safe range with hysteresis.

Temperature limits
Actual value: 21.8 °C

High limit [°C]	Low limit [°C]	Hysteresis [°C]	Time delay [sec]
50.0	0.0	1.0	30

Relative humidity limits
Actual value: 66.3%RH

High limit [%RH]	Low limit [%RH]	Hysteresis [%RH]	Time delay [sec]
80.0	20.0	1.0	30

Dew point limits
Actual value: 15.3°C

High limit [°C]	Low limit [°C]	Hysteresis [°C]	Time delay [sec]
50.0	0.0	1.0	30

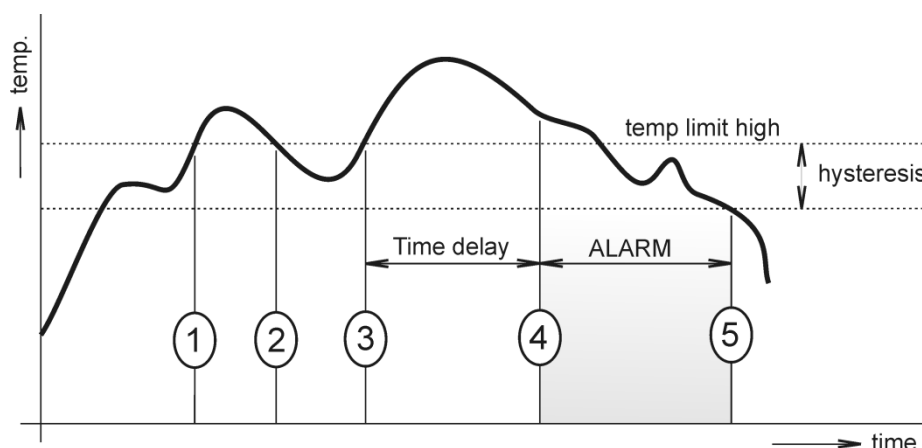
Atmospheric pressure limits
Actual value: 96.09kPa

High limit [kPa]	Low limit [kPa]	Hysteresis [kPa]	Time delay [sec]
100.00	70.00	1.00	30

Apply settings

Cancel changes

Example of setting the limit to the upper alarm limit:



In Point 1 the temperature exceeded the limit. From this time, the time-delay is counting. Because at point 2 the temperature dropped below the limit value before the time delay expired, alarm was not set.

In Point 3 the temperature has risen over limit again. During the time-delay the value does not drop below the set limit, and therefore was in Point 4 caused alarm. At this moment were sent e-mails, traps and set alarm flag on website, SNMP and Modbus.

The alarm lasted up to Point 5, when the temperature dropped below the set hysteresis (temperature limit – hysteresis). At this moment was active alarm cleared and e-mail sent.

When alarm occurs, alarm messages will be sent. In case of power failure or device reset (e.g. changing the configuration) will new alarm state evaluated and new alarm messages will be sent.

Measuring

Page contains settings of the measuring part of the device. Settings can be saved only if jumper inside device is closed. Fields in this page are shown according device type.

It is possible set visibility of measured values on device LCD. Display can be switched off. Units of measured and computed quantities are selectable by fields **Temperature unit**, **Computed value** and **Atmospheric pressure unit**. Devices without atmospheric pressure measurement have field **Pressure value for computed values**. Here is required to set pressure for calculation of computed quantities. Pressure compensation for altitude above sea-level is possible via **Atmospheric pressure offset**. This offset depends on current altitude. This value is added to measured atmospheric pressure, final value must be in range 300hPa to 1350hPa.

Back
Exit to main menu

Settings

Measuring

General
General device settings

Network
Basic settings of the network interface

Alarm limits
Configuration of the alarm limits

Measuring
Units, display settings, etc.

SOAP protocol
Setup SOAP protocol for database system

Email
Alarm emails configuration

Protocols
Syslog and ModbusTCP protocol settings

SNMP
SNMPv1 protocol and SNMP Traps

Time
Synchronization with NTP server

WWW and Security
Web server and Security configuration

MinMax memory
Timestamps, clear memory

Maintenance
Factory defaults, info, etc.

Display
Display can be configured to show required values. It is recommended to switch off display when device is used in temperature over 70°C.

LCD enabled☒

Show Temperature☒

Show Relative humidity☒

Show Computed value☒

Show Atmospheric pressure☒

Units
Device measures values according selected physical units. History graphs and tables will be cleared after changing units.

Temperature Unit°C

Computed valueDew point

Atmospheric pressure UnitkPa

Other settings
Compensation of measured values and other settings. For more information read the user guide carefully.

Pressure value for computed values1013 [hPa]

Atmospheric pressure offset0.0 [hPa]

Apply settings

Cancel changes

ATAL

13

SOAP protocol

SOAP protocol can be enabled by option **SOAP protocol enabled**. Destination SOAP server can be set via **SOAP server address**. For setup of server port can be used option **SOAP server port**. Device sends SOAP message according selected **Sending interval**. Option **Send SOAP message when alarm occurs** sends message when an alarm on channel occurs or alarm is cleared. These SOAP messages are sent asynchronously to selected interval.

Back
Exit to main menu

Settings SOAP

General
General device settings

Network
Basic settings of the network interface

Alarm limits
Configuration of the alarm limits

Measuring
Units, display settings, etc.

SOAP protocol
Setup SOAP protocol for database system

Email
Alarm emails configuration

Protocols
Syslog and ModbusTCP protocol settings

SNMP
SNMPv1 protocol and SNMP Traps

Time
Synchronization with NTP server

WWW and Security
Web server and Security configuration

MinMax memory
Timestamps, clear memory

Maintenance
Factory defaults, info, etc.

Setup SOAP protocol for database system. Current measured values are sent as XML files. For more information please read the user guide for database system.

SOAP protocol enabled☒

SOAP server address

SOAP server port

Sending interval

Send SOAP message when alarm occurs☒

Apply settings

Cancel changes

14

ATAL

Email

Email sending enabled option allows email features. It is necessary set address of the SMTP server into **SMTP server address** field. Domain name for SMTP server can be used. Default port of the SMTP server can be changed using item **SMTP server port**. SMTP authentication can be enabled using **SMTP authentication** option. When authentication is enabled **Username** and **Password** must be set.

For successfully email sending it is necessary insert **Email sender address**. This address is usually same as username of the SMTP authentication. Into fields **Recipient 1** to **Recipient 3** it is possible set address of email recipients. Option **Short email** enable sending emails in short format. This format is usable when you need to forward emails into SMS messages.

When option **Alarm email repeat sending interval** is enabled and there is active alarm on channel, then emails with actual values are sent repeatedly. **Info email sending interval** option enables sending emails at selected time interval. CSV history file can be sent together with the repeat/info emails. This feature can be enabled by **Alarm and Info emails attachment** option.

It is possible to test email function using button **Apply and test**. This button save a new settings and send a testing email immediately.

Back
Exit to main menu

Settings

General
General device settings

Network
Basic settings of the network interface

Alarm limits
Configuration of the alarm limits

Measuring
Units, display settings, etc.

SOAP protocol
Setup SOAP protocol for database system

Email
Alarm emails configuration

Protocols
Syslog and ModbusTCP protocol settings

SNMP
SNMPv1 protocol and SNMP Traps

Time
Synchronization with NTP server

WWW and Security
Web server and Security configuration

MinMax memory
Timestamps, clear memory

Maintenance
Factory defaults, info, etc.

Email

Configuration of the alarm emails. Device can send warning email when alarm on measured channel occurs. Email is also sent when alarm condition is cleared.

Email sending enabled ☒

SMTP server configuration
For proper email sending it is necessary to setup connection to your SMTP server. SMTP authentication can be used if needed. For information about SMTP server settings please contact your network administrator or ISP.

SMTP server address

SMTP server port

SMTP authentication ☒

Username

Password

Email configuration
Setup up to three address for email recipients. Sender address is usually the same as the username of the SMTP authentication.

Email sender address

Recipient 1

Recipient 2

Recipient 3

Short email ☐

Alarm email repeat sending interval

Info email sending interval

Alarm and Info emails attachment

Apply settings

Apply and test

Cancel changes

Modbus a Syslog protocols

ModbusTCP and Syslog protocol settings are configurable via menu **Protocols**. Modbus server is enabled by default. Deactivation is possible via **Modbus server enabled** option. Modbus port can be changed via **Modbus port** field. Syslog protocol can be enabled using item **Syslog enabled**. Syslog messages are sent to IP address of the Syslog server - field **Syslog server IP address**.

Back
Exit to main menu

Settings

Protocols

General
General device settings

Network
Basic settings of the network interface

Alarm limits
Configuration of the alarm limits

Measuring
Units, display settings, etc.

SOAP protocol
Setup SOAP protocol for database system

Email
Alarm emails configuration

Protocols
Syslog and ModbusTCP protocol settings

SNMP
SNMPv1 protocol and SNMP Traps

Time
Synchronization with NTP server

WWW and Security
Web server and Security configuration

MinMax memory
Timestamps, clear memory

Maintenance
Factory defaults, info, etc.

ModbusTCP protocol
Configuration of the ModbusTCP protocol. Actual measured values can be read using ModbusTCP protocol. Modbus registers are described inside the user manual.

Modbus server enabled ☒

Modbus port

Syslog protocol
Configuration of the Syslog protocol. Alarm messages can be sent to the Syslog server using UDP protocol.

Syslog enabled ☒

Syslog server IP address

Apply settings

Cancel changes

SNMP

For reading values via SNMP it is necessary to know password - **SNMP read community**. SNMP Trap can be delivered up to three IP address - **IP address of the Trap recipient**. SNMP Traps are sent at alarm or error state on the channel. Trap feature can be enabled by option **Trap enabled**.

Back
Exit to main menu

Settings

SNMP

General
General device settings

Network
Basic settings of the network interface

Alarm limits
Configuration of the alarm limits

Measuring
Units, display settings, etc.

SOAP protocol
Setup SOAP protocol for database system

Email
Alarm emails configuration

Protocols
Syslog and ModbusTCP protocol settings

SNMP
SNMPv1 protocol and SNMP Traps

Time
Synchronization with NTP server

WWW and Security
Web server and Security configuration

MinMax memory
Timestamps, clear memory

Maintenance
Factory defaults, info, etc.

Configuration of the SNMPv1 protocol and SNMP Traps. Actual measured values can be read using SNMP protocol. When alarm on channel occurs a warning message (Trap) can be sent to selected IP addresses.

SNMP read community

Trap enabled ☒

IP address of the Trap recipient 1

IP address of the Trap recipient 2

IP address of the Trap recipient 3

Time

Time synchronization with SNTP server can be enabled by **Time synchronization enabled** option. IP address of the SNTP is necessary to set into **SNTP server IP address** item. SNTP time is synchronized at UTC format, and due to be necessary set corresponding time offset - **GSM offset [min]**. Time is synchronized every 24 hours by default. Option **NTP synchronization every hour** decrease this synchronization interval to one hour.

Back
Exit to main menu

Settings

Time

General
General device settings

Network
Basic settings of the network interface

Alarm limits
Configuration of the alarm limits

Measuring
Units, display settings, etc.

SOAP protocol
Setup SOAP protocol for database system

Email
Alarm emails configuration

Protocols
Syslog and ModbusTCP protocol settings

SNMP
SNMPv1 protocol and SNMP Traps

Time
Synchronization with NTP server

WWW and Security
Web server and Security configuration

MinMax memory
Timestamps, clear memory

Maintenance
Factory defaults, info, etc.

Time can be synchronized according the SNTP server. To correct time is necessary set GMT offset of your time zone.

Time synchronization enabled☒

SNTP server IP address

GMT offset [min]

NTP synchronization every hour☐

Apply settings

Cancel changes

18

ATAL

WWW and security

Security features can be enabled by the **Security enabled** option. When security is enabled it is necessary to set administrator password. This password will be required for device settings. When secured access is required even to actual values reading it is possible to enable **User account only for viewing enabled**. Port of the www server can be changed from the default value 80 using field **WWW port**. Web pages with actual values are refreshed according to **Web refresh interval** field.

Back
Exit to main menu

Settings

WWW and Security

General
General device settings

Network
Basic settings of the network interface

Alarm limits
Configuration of the alarm limits

Measuring
Units, display settings, etc.

SOAP protocol
Setup SOAP protocol for database system

Email
Alarm emails configuration

Protocols
Syslog and ModbusTCP protocol settings

SNMP
SNMPv1 protocol and SNMP Traps

Time
Synchronization with NTP server

WWW and Security
Web server and Security configuration

MinMax memory
Timestamps, clear memory

Maintenance
Factory defaults, info, etc.

Security
Configuration of the secure access to the device. Administrator password must be inserted if security is enabled. Administrator password is used for device configuration. User password is used only for measured values viewing.

Security enabled☒

Administrator username

Administrator password

Confirm Administrator password

User account only for viewing enabled☒

User username

User password

Confirm User password

Web server
Configuration of the embedded web server. After web port changing, device will be rebooted with new port.

Web server enabled☒

WWW port

Web refresh interval

Apply settings

Cancel changes

Setup using TSensor software

TSensor software is an alternative to web configuration. Some less important parameters are configurable only by the TSensor software. TSensor software allows user adjustment of measured values.

Parameter **MTU size** can reduce size of the Ethernet frame. Lowering of this size can solve some communication problems mainly with Cisco network infrastructure.

ATAL

19

Factory defaults

Factory defaults field set the device into factory configuration. Network parameters (IP address, Subnet mask, Gateway, DNS) are left without changes. Settings relating to measurement are not restored by factory defaults.

[Back](#)
Exit to main menu

Settings

Maintenance

General
General device settings

Network
Basic settings of the network interface

Alarm limits
Configuration of the alarm limits

Measuring
Units, display settings, etc.

SOAP protocol
Setup SOAP protocol for database system

Email
Alarm emails configuration

Protocols
Syslog and ModbusTCP protocol settings

SNMP
SNMPv1 protocol and SNMP Traps

Time
Synchronization with NTP server

WWW and Security
Web server and Security configuration

MinMax memory
Timestamps, clear memory

Maintenance
Factory defaults, info, etc.

Info
Basic informations about device. Find more detailed information on the diagnostic page. Please send the diagnostic file together with request to support.

Device typeT7511

Serial number11963408

MAC address00-20-4A-A6-E7-BD

Firmware version1-5-5-0.0978 / 1.00

Build firmware noticeRTM

Device uptime4 h, 29 min, 28 sec

Diagnostic file[192.168.1.41/diag.log](#)

Restart
Device will be restarted after dialog confirmation. All history values are cleared after restart. Restarting of the device will take a few seconds.

Restart device

Factory defaults
Factory defaults button restores device to factory settings. Network parameters like a IP address, subnet mask and gateway IP will not be changed.

Factory defaults

Network parameters are changed while you close jumper and push button inside device during connecting power supply. Factory parameters settings:

Parameter	Default value
SMTP server address	example.com
SMTP server port	25
Alarm email repeat sending interval	off
Info email repeat sending interval	off
Alarm and Info emails attachment	off
E-mail recipients addresses	cleared
E-mail sender	sensor@IP
SMTP authentication	off
SMTP user/SMTP password	cleared
E-mail sending enabled	off
IP addresses SNMP traps recipients	0.0.0.0
Password for SNMP reading	public
Sending SNMP Trap	off
Website refresh interval [sec]	10
Website enabled	yes
Website port	80
Security	off

Administrator password		cleared
User password		cleared
Modbus port		502
Modbus server enabled		yes
History storage interval [sec]		60
SOAP server port		80
SOAP server address		cleared
SOAP sending interval [sec]		60
SOAP message when alarm occurs		yes
SOAP protocol enabled		off
Syslog server IP address		0.0.0.0
Syslog protocol enabled		off
SNTP server IP address		217.31.205.226
GMT offset [min]		0
NTP synchronization every hour		off
SNTP synchronization enabled		off
MTU		1400
Periodic restart interval		off
Demo mode		off
Temperature	High limit	50
	Low limit	0
	Hysteresis	1
	Time delay [sec]	30
Relative humidity	High limit	80
	Low limit	20
	Hysteresis	1
	Time delay [sec]	30
Computed quantity	High limit	50
	Low limit	0
	Hysteresis	1
	Time delay [sec]	30
Atmospheric pressure	High limit	1000
	Low limit	700
	Hysteresis	12
	Time delay [sec]	30
Device name		Web Sensor

Communication protocols

Short introduction to communication protocols of the device. To use some communication protocols is necessary software, which can use the protocol. This software is not included. For detailed description of protocols and application notes please contact your distributor.

Website

The device supports displaying of measured values, history graphs and configuration using web browser. History graphs are based on HTML5 canvas. Web browser must support this feature for proper function of graphs. Firefox, Opera, Chrome or Internet Explorer 9 can be used. If the device has IP address **192.168.1.213** type into your browser **http://192.168.1.213**. Using TSensor software or web interface can be set automatic webpages refresh in interval. The default value is 10sec. Actual measured values can be obtained using XML file **values.xml**.

Values from history can be exported in CSV format. History storage interval can be set using TSensor software or web interface. History is erased after every reboot of the device. Reboot of the device is performed when the power supply is disconnected and also after configuration change.

The device allows you to customize the design of web pages. For detailed information please contact your distributor.

SMTP – sending e-mails

When measured values are over the set limits, the device allows send e-mail to a maximum of 3 addresses. E-mail is send when alarm condition on the channel is cleared or a measuring error occurs. It is possible to set repeat interval for email sending. For correct sending of e-mails it is necessary to set address of SMTP server. Domain address can be used as SMTP server address too. For proper function of DNS is required to set DNS server IP address. SMTP authentication is supported but SSL/STARTTLS not. Standard SMTP port 25 is used by default. SMTP port can be changed. Contact your network administrator to obtain configuration parameters of your SMTP server. E-mail sent by the device cannot be answered.

SNMP

Using SNMP protocol you can read actual measured values, alarm status and alarm parameters. Via SNMP protocol is also possible to get last 1000 measured values from history table. Writing via SNMP protocol is not supported. It is supported **SNMPv1** protocol version only. SNMP used **UDP port 161**. OID keys description can be found in the MIB table, which can be obtained from device website or from your distributor. The

password for reading is factory set to **public**. The changes can be made using TSensor software or web interface. OID keys:

OID	Description	Type
.1.3.6.1.4.1.22626.1.2.1		
.1.3.6.1.4.1.22626.1.2.1.1	Actual measured temperature	String
.1.3.6.1.4.1.22626.1.2.1.2	Actual measured relative humidity	String
.1.3.6.1.4.1.22626.1.2.1.3	Actual measured computed quantity	String
.1.3.6.1.4.1.22626.1.2.1.4	Actual measured atmospheric pressure	String
.1.3.6.1.4.1.22626.1.2.1.5	Temperature alarm state ("none", "high", "low")	String
.1.3.6.1.4.1.22626.1.2.1.6	Relative humidity alarm state	String
.1.3.6.1.4.1.22626.1.2.1.7	Computed quantity alarm state	String
.1.3.6.1.4.1.22626.1.2.1.8	Atmospheric pressure alarm state	String
.1.3.6.1.4.1.22626.1.2.1.9	Temperature unit	String
.1.3.6.1.4.1.22626.1.2.1.10	Relative humidity unit	String
.1.3.6.1.4.1.22626.1.2.1.11	Computed quantity unit	String
.1.3.6.1.4.1.22626.1.2.1.12	Atmospheric pressure unit	String
.1.3.6.1.4.1.22626.1.2.1.13	Min. temperature memory	String
.1.3.6.1.4.1.22626.1.2.1.14	Min. relative humidity memory	String
.1.3.6.1.4.1.22626.1.2.1.15	Min. computed value memory	String
.1.3.6.1.4.1.22626.1.2.1.16	Min. atmospheric pressure memory	String
.1.3.6.1.4.1.22626.1.2.1.17	Max. temperature memory	String
.1.3.6.1.4.1.22626.1.2.1.18	Max. relative humidity memory	String
.1.3.6.1.4.1.22626.1.2.1.19	Max. computed value memory	String
.1.3.6.1.4.1.22626.1.2.1.20	Max. atmospheric pressure memory	String
.1.3.6.1.4.1.22626.1.2.2		
.1.3.6.1.4.1.22626.1.2.2.1	Device name	String
.1.3.6.1.4.1.22626.1.2.2.2	Device serial number	String
.1.3.6.1.4.1.22626.1.2.2.3	Device type	String
.1.3.6.1.4.1.22626.1.2.3		
.1.3.6.1.4.1.22626.1.2.3.1	Actual measured temperature	Int*10
.1.3.6.1.4.1.22626.1.2.3.2	Actual measured relative humidity	Int*10
.1.3.6.1.4.1.22626.1.2.3.3	Actual measured computed quantity	Int*10
.1.3.6.1.4.1.22626.1.2.3.4	Actual measured atmospheric pressure	Int*X
.1.3.6.1.4.1.22626.1.2.3.5	Temperature alarm (0–none, 1–high, 2–low)	Integer
.1.3.6.1.4.1.22626.1.2.3.6	Relative humidity alarm state	Integer
.1.3.6.1.4.1.22626.1.2.3.7	Computed quantity alarm state	Integer
.1.3.6.1.4.1.22626.1.2.3.8	Atmospheric pressure alarm state	Integer
.1.3.6.1.4.1.22626.1.2.4		
.1.3.6.1.4.1.22626.1.2.4.1	Temperature low limit	Int*10
.1.3.6.1.4.1.22626.1.2.4.2	Temperature high limit	Int*10
.1.3.6.1.4.1.22626.1.2.4.3	Relative humidity low limit	Int*10
.1.3.6.1.4.1.22626.1.2.4.4	Relative humidity high limit	Int*10
.1.3.6.1.4.1.22626.1.2.4.5	Computed quantity low limit	Int*10
.1.3.6.1.4.1.22626.1.2.4.6	Computed quantity high limit	Int*10
.1.3.6.1.4.1.22626.1.2.4.7	Temperature time-delay	Integer
.1.3.6.1.4.1.22626.1.2.4.8	Relative humidity time-delay	Integer

.1.3.6.1.4.1.22626.1.2.4.9	Computed quantity time-delay	Integer
.1.3.6.1.4.1.22626.1.2.4.10	Temperature hysteresis	Int*10
.1.3.6.1.4.1.22626.1.2.4.11	Relative humidity hysteresis	Int*10
.1.3.6.1.4.1.22626.1.2.4.12	Computed quantity hysteresis	Int*10
.1.3.6.1.4.1.22626.1.2.4.13	Atmospheric pressure low limit	Int*X
.1.3.6.1.4.1.22626.1.2.4.14	Atmospheric pressure high limit	Int*X
.1.3.6.1.4.1.22626.1.2.4.15	Atmospheric pressure time-delay	Integer
.1.3.6.1.4.1.22626.1.2.4.16	Atmospheric pressure hysteresis	Int*X
.1.3.6.1.4.1.22626.1.5.5.1.0	SNMP Trap	String
.1.3.6.1.4.1.22626.1.5.6.1.1.1.nr	Temperature history values	Int*10
.1.3.6.1.4.1.22626.1.5.6.1.1.2.nr	Relative humidity history values	Int*10
.1.3.6.1.4.1.22626.1.5.6.1.1.3.nr	Computed quantity history values	Int*10
.1.3.6.1.4.1.22626.1.5.6.1.1.4.nr	Atmospheric pressure history values	Int*X

When alarm occurred a warning messages (trap) can be sent to selected IP addresses. Addresses can be set using TSensor software or web interface. Traps are sent via **UDP** protocol on **port 162**. The device can send following traps:

Trap	Description	
0/0	Reset of the device	
1/0	Settings was changed	
6/0	Testing Trap	
6/1	NTP synchronization error	
6/2	E-mail sending error	SMTP server login error
6/3		SMTP authentication error
6/4		Some error occurred during SMTP communication
6/5		TCP connection to server cannot be opened
6/6		DNS error
6/7	SOAP message sending error	SOAP file not found inside web memory
6/8		DNS error or Host error
6/9		TCP connection to server cannot be opened
6/10		Wrong response code from the SOAP server
6/11 – 6/14	Upper alarm on channel	
6/21 – 6/24	Lower alarm on channel	
6/31 – 6/34	Clearing alarm on channel	
6/41 – 6/44	Measuring error	

Modbus TCP

Device supports Modbus protocol for communication with SCADA systems. Device use Modbus TCP protocol. **TCP port** is set to **502** by default. Port can be changed using TSensor software or web interface. Only two Modbus clients can be connected to device at one moment. Modbus device address (Unit Identifier) can be arbitrary. Modbus write command is not supported. Specification and description of the Modbus protocol is free to download on: www.modbus.org.

Supported Modbus commands (functions):

Command	Code	Description
Read Holding Register (s)	0x03	Read 16b register(s)

Modbus device registers. Address could be by 1 higher, depending on type used communication library:

Address [DEC]	Address [HEX]	Value	Type
49	0x0031	Measured temperature	Int*10
50	0x0032	Measured relative humidity	Int*10
51	0x0033	Measured computed quantity	Int*10
52	0x0034	Measured atmospheric pressure	Int*X
53	0x0035	Dew point temperature	Int*10
54	0x0036	Absolute humidity	Int*10
55	0x0037	Specific humidity	Int*10
56	0x0038	Mixing ration	Int*10
57	0x0039	Specific enthalpy	Int*10
4149	0x1035	Serial number high	BCD
4150	0x1036	Serial number low	BCD
4151	0x1007	Device type	HEX
20481	0x5001	Temperature low limit	Int*10
20482	0x5002	Temperature high limit	Int*10
20483	0x5003	Relative humidity low limit	Int*10
20484	0x5004	Relative humidity high limit	Int*10
20485	0x5005	Computed quantity low limit	Int*10
20486	0x5006	Computed quantity high limit	Int*10
20487	0x5007	Temperature hysteresis	Int*10
20488	0x5008	Temperature time-delay	DEC
20489	0x5009	Relative humidity hysteresis	Int*10
20490	0x500A	Relative humidity time-delay	DEC
20491	0x500B	Computed quantity hysteresis	Int*10
20492	0x500C	Computed quantity time-delay	DEC
20493	0x500D	Temperature alarm state	String
20494	0x500E	Relative humidity alarm state	String
20495	0x500F	Computed quantity alarm state	String
20496	0x5010	Atmospheric pressure low limit	Int*X
20497	0x5011	Atmospheric pressure high limit	Int*X
20498	0x5012	Atmospheric pressure hysteresis	Int*X
20499	0x5013	Atmospheric pressure alarm state	String
20500	0x5014	Atmospheric pressure time-delay	DEC
20501	0x5015	Min. temperature memory	Int*10

20502	0x5016	Max. temperature memory	Int*10
20503	0x5017	Min. relative humidity memory	Int*10
20504	0x5018	Max. relative humidity memory	Int*10
20505	0x5019	Min. computed value memory	Int*10
20506	0x501A	Max. computed value memory	Int*10
20507	0x501B	Min. atmospheric pressure memory	Int*X
20508	0x501C	Max. atmospheric pressure memory	Int*X

Type:

DEC	register is in range 0 – 4500 (16bit)
BCD	register is in BCD format (16bit)
HEX	number in HEX format (16bit)
String	two characters in one 16bit register: no – no alarm hi – value is lower than set limit lo – value is higher than set limit
Int*10	register is in format integer*10 – 16 bits: (125=12.5°C; error = 9999 or -9999)
Int*X	register format depends on value type (error = -9999): hPa – integer*10 (9760 = 976.0hPa) PSI – integer*1000 (14156 = 14.156PSI) inHg – integer*100 (2882 = 28.82inHg) mBar – integer*10 (9761 = 976.1mBar) oz/in ² – integer*10 (2265 = 226.5oz/in ²) mmHg – integer*10 (7321 = 732.1mmHg) inH ₂ O – integer*10 (3919 = 391.9inH ₂ O) kPa – integer*100 (9761 = 97.61kPa)

SOAP

The device allows you to send currently measured values via **SOAP v1.1** protocol. The device sends values in XML format to the web server. The advantage of this protocol is that communication is initialized by the device side. Due to it is not necessary use port forwarding. If the SOAP message cannot be delivered, warning message via SNMP Trap or Syslog protocol is sent. SOAP message example:

```
<?xml version="1.0" encoding="utf-8"?>
<soap:Envelope xmlns:soap="http://schemas.xmlsoap.org/soap/envelope/"
  xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
  xmlns:xsd="http://www.w3.org/2001/XMLSchema">
  <soap:Body>
    <InsertTx5xxSample xmlns="http://cometsystem.cz/schemas/soapTx5xx_v2.xsd">
      <passKey>13960932</passKey>
      <device>4145</device>
```

```

    <temp>1.4</temp>
    <relHum>91.9</relHum>
    <compQuant>0.3</compQuant>
    <pressure>-9999</pressure>
    <alarms>hi,no,no,no</alarms>
    <compType>Dew point</compType>
    <tempU>C</tempU>
    <pressureU>n/a</pressureU>
    <timer>60</timer>
  </InsertTx5xxSample>
</soap:Body>
</soap:Envelope>

```

Element	Description
<passKey>	Contains the device serial number (an eight digit number).
<device>	Device type identification number (code):
	Device
	Code[DEC]
	EPD-TRH
	4107
	EPD-T
	4106
	EPD-TRHP
	4129
<temp>	EPD-P-INT
	4124
	EPD-T-INT
	4144
	EPD-TRH-INT
	4145
	EPD-TRHP-INT
	4146
<relHum>	Contains the value of relative humidity. Error value: 9999 or -9999 .
<compQuant>	Contains the value of computed quantity. Error value: 9999 or -9999 .
<pressure>	Contains the value of atmospheric pressure. Error value: -9999 .
<alarms>	State of alarm on temperature, relative humidity, computed quantity and pressure channel. Format: tm,rh,cq,pr where: tm – temperature alarm, rh – relative humidity alarm, cq – computed quantity alarm, pr – atmospheric pressure alarm and alarm values: no – no alarm or value is not supported, hi – high alarm, lo – low alarm.
<compType>	Computed quantity type: Absolute humidity, Specific humidity, Mixing proportion, Specific enthalpy, Dew point or n/a .
<tempU>	Temperature and dew point unit. Values: C – temperature in °C, F – temperature in °F.
<pressureU>	Atmospheric pressure unit. Values: hPa, PSI, inHg, mBar, oz/in^2, mmHg, inH2O and kPa .
<timer>	SOAP sending interval [sec].

Example of the web service for incoming SOAP messages from Web Sensor Tx5xx. Example uses Apache (2.2.10) web server and PHP (5.2.6). It is required to install and enable SOAP extension for PHP. This example store incoming messages to the hard drive:

<?

```
function InsertTx5xxSample($passKey,$device,$temp,$relHum,$compQuant,$pressure,
```

```

        $alarms,$compType,$tempU,$pressureU,$timer) {
    $data = "Time: ".StrFTime("%y/%m/%d %H:%M:%S", Time()).", Temp: ".$temp.
        ", RH: ".$relHum.", CQ: ".$compQuant. ", Pressure: ".$pressure."\n";
    $file_write = FOpen("soap.log", "a");
    FWrite($file_write, $data);
    FClose($file_write);
}
$server = new SoapServer(null, array('uri' => "http://test-uri/"));
$server->addFunction('InsertTx5xxSample');
$server->handle();
?>

```

Syslog

The device allows sending text message to selected Syslog server. Events are send using **UDP** protocol on **port 514**. Syslog protocol implantation is according to RFC5424 and RFC5426. Events when Syslog messages are send:

| Text | Event |
|---|----------------------------|
| Sensor - fw 1-5-5.x | Reset of the device |
| Settings changed | Settings was changed |
| NTP synchronization error | NTP synchronization error |
| Testing message | Test Syslog message |
| Email login error | E-mail sending error |
| Email auth error | |
| Email some error | |
| Email socket error | |
| Email dns error | |
| SOAP file not found | |
| SOAP host error | SOAP message sending error |
| SOAP sock error | |
| SOAP delivery error | |
| SOAP dns error | |
| Text XXXX is set according channel type: Temperature, Relative humidity, Specific humidity, Mixing proportion, Specific enthalpy, Atmospheric pressure. | |
| High alarm XXXX | Upper alarm on channel |
| Low alarm XXXX | Lower alarm on channel |
| Clearing XXXX alarm | Clearing alarm on channel |
| Error XXXX | Measuring error |

SNTP

The device allows time synchronization with NTP (SNTP) server. SNMP protocol version 3.0 is supported (RFC1305). Time synchronization is made every 24 hours. Time synchronization every hour can be enabled. For time synchronization it is necessary set IP address to the SNTP server. It is also possible set GMT offset for correct time zone. Time is used in graphs and history CSV files. Maximum jitter between two time synchronization is 90sec at 24 hours interval.

Troubleshooting

The chapter describes the common problems with devices Web Sensor Tx5xx and methods how to fix these problems. Please read this chapter before you will call technical support.

I forgot the device IP address

IP address is factory set to **192.168.1.213**. If you had changed it and forgot new IP address, run the TSensor software and press **Find device...** In the window are displayed all available devices.

I cannot connect to the device

In search window is only IP and MAC address displayed

Other details are marked **N/A**. This problem occurs if IP address of the device is set to another network.

Select the window **Find device** in TSensor software and press **Change IP address**. Follow the software instructions. To assign IP address automatically using DHCP server, set the device IP address to **0.0.0.0**.

Device IP address is not displayed in window Find device

In TSensor software menu press **Help! My device was not found!** in window **Find device**. Follow the software instructions. MAC address of the device can be found on product label.

The device is not found even after manually setting MAC address

This problem occurs especially in cases when the IP address of the device belongs to another network and also Subnet mask or Gateway are incorrect.

In this case is DHCP server in the network necessary. In TSensor software menu press **Help! My device was not found!** in window **Find device**. As new IP address set **0.0.0.0**. Follow the software instructions. An alternative is to reset device to factory defaults using factory-defaults button.

I forgot the password for setup

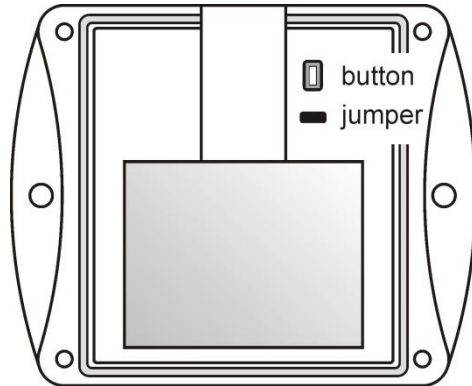
Please reset device to factory defaults. Procedure is described at following point.

Factory defaults

This procedure restore device to factory settings including network parameters (IP address, Subnet mask, etc.). Settings relating to measurement are not restored by factory defaults. For factory-defaults follow these steps:

- disconnect the power supply

- unscrew upper cover of the device case
- close the jumper inside device
- press button inside device and power on device at same time
- keep the button pressed for 10 sec



- close the device

Display is off

Check if power supply is connected. Disconnect power connector and connect connector again – watch the display at the moment of connecting the power. If all LCD segments light for 1 second and go out again, the display is turned off by software.

Device error states

Device still make self-tests. If error occurred, LCD shows error codes:

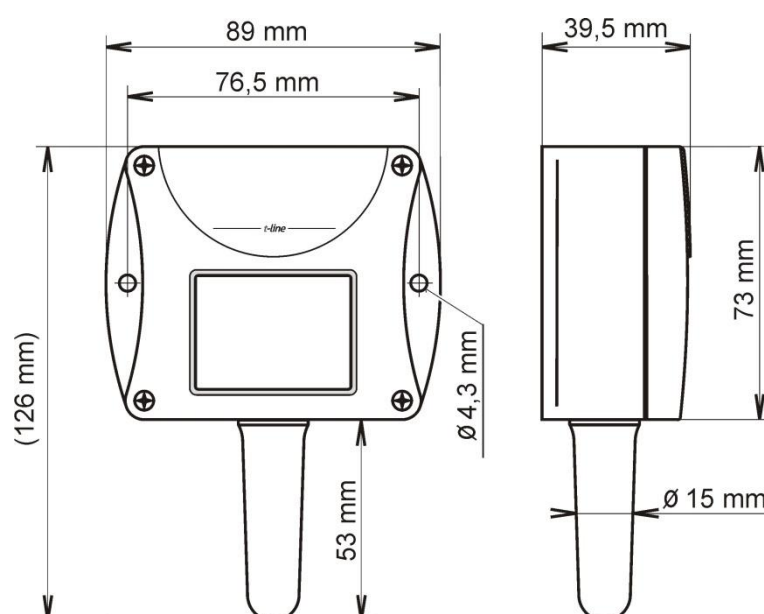
| Error | LCD display | Description |
|---------|-------------|--|
| Error 0 | Err0 | Internal memory CRC error. In this state device doesn't work. This is a critical error, contact the distributor. |
| Error 1 | Err1 | Measured or computed quantity is over the upper limit. Error code 9999. This state appears in case of: <ul style="list-style-type: none">- Measured temperature is higher than approximately +600°C (i.e. high non-measurable resistance of temperature sensor, probably open circuit)- Relative humidity is higher than 100%RH (i.e. damaged humidity sensor or humidity calculation is not possible due to temperature error)- Unable to calculate computed quantity (temperature or humidity measurement error) |
| Error 2 | Err2 | Measured or computed quantity is under lower limit or is error in pressure measuring. Error code -9999. This state appears in case of: <ul style="list-style-type: none">- Measured temperature is lower than approximately -210°C (i.e. resistance is too small, probably short circuit)- Relative humidity is lower than 0%RH (i.e. damaged humidity sensor or humidity calculation is not possible due to temperature error)- Measured atmospheric pressure with offset is over range 300hPa to 1350hPa or the pressure sensor is damaged- Unable to calculate computed quantity (temperature or humidity measurement error) |
| Error 3 | Err3 | Internal A/D converter error. Error code -9999. In this state device doesn't work. Contact the distributor. |
| Error 4 | Err4 | Internal pressure sensor error. Error code -9999. In this state device not measure pressure. Contact the distributor. |

Technical specifications

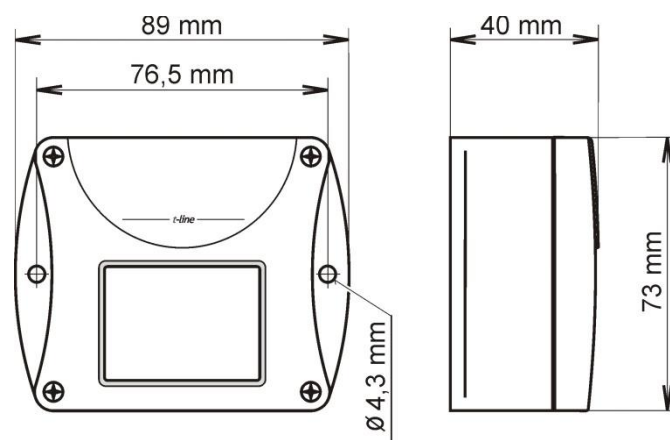
Information about technical specifications of the device.

Dimensions

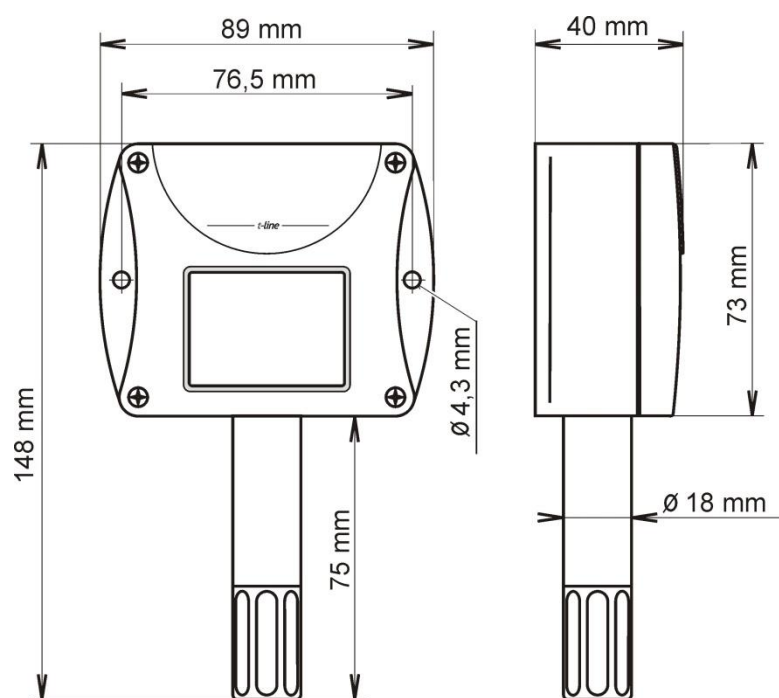
Web Sensor EPD-T-INT:



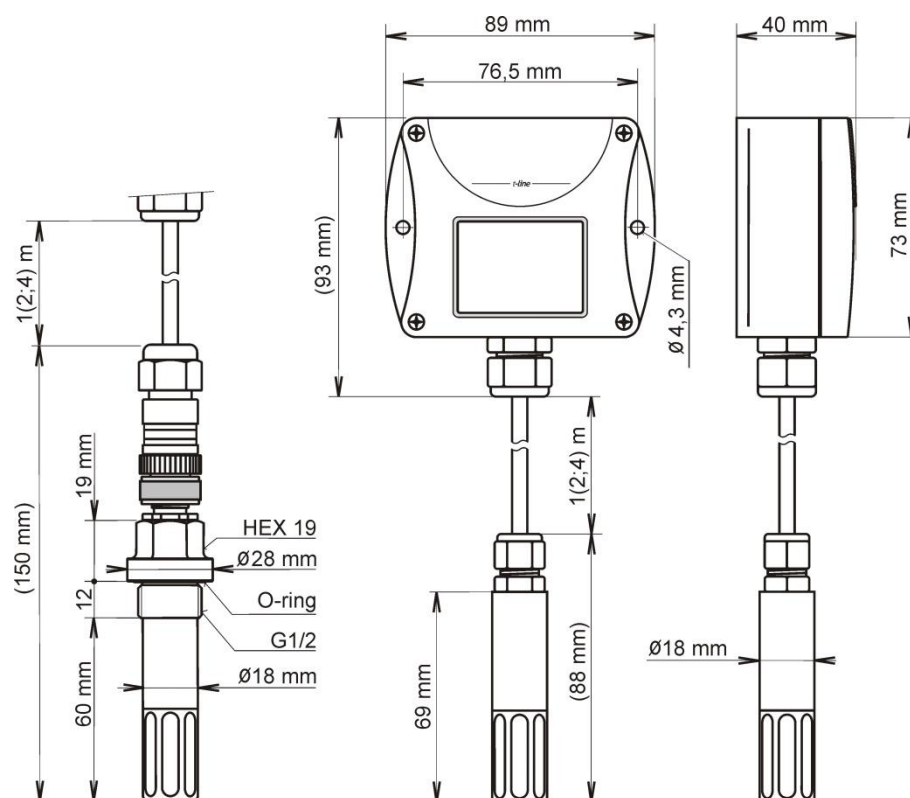
Web Sensor EPD-P-INT:



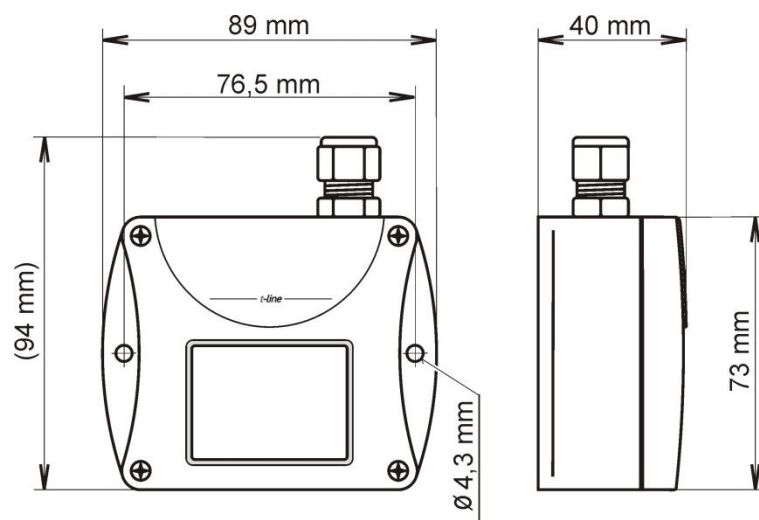
Web Sensor EPD-TRH-INT, EPD-TRHP-INT:



Web Sensor EPD-TRH(P), EPD-TRHP:



Web Sensor EPD-T:



General parameters

Supply voltage:

DC voltage from 9V to 30V, coaxial connector, 5x2.1mm diameter, positive pole in the middle

Consumption:

~ 1W depending on the operating mode

Measuring interval:

2sec

Display switching interval:

4sec (when more than two values are displayed)

Not allowed manipulations:

It is not allowed to operate the device under conditions other than specified in technical parameters. Devices are not designed for locations with chemically aggressive environment. Temperature and humidity sensors must not be exposed to direct contact with water or other liquids. It is not allowed to remove the sensor cover to avoid any mechanical damage of the sensors.

Case material:

ABS

Material of EPD-TRHP probe:

duralumin with the black eloxal surface finish

Mechanical connection of probe EPD-TRHP:

G1/2 with O-ring

Weight:

EPD-P-INT ~130g

EPD-T-INT ~145g

EPD-T ~145g

EPD-TRH-INT ~155g

EPD-TRHP-INT ~155g

EPD-TRH/1m probe ~210g, EPD-TRH/2m probe ~250g, EPD-TRH/4m probe ~330g

EPD-TRHP/1m probe ~260g, EPD-TRHP/2m probe ~300g, EPD-TRHP/4m probe ~380g

EPD-TRHP/1m probe ~210g, EPD-TRHP/2m probe ~250g, EPD-TRHP/4m probe ~330g

Mounting the device:

With two holes at the bottom of the unit

Communication port:

RJ45 connector, 10Base-T/100Base-TX Ethernet (Auto-Sensing)

Recommended Connector Cable:

for industrial use is recommended Cat5e STP cable, in less demanding applications can be replaced by Cat5 cable, maximum cable length 100m

Supported protocols:

TCP/IP, UDP/IP, ARP, ICMP, DHCP, TFTP, DNS

HTTP, SMTP, SNMPv1, ModbusTCP, SNMP, SOAPv1.1, Syslog

Supported SNMP authentication types:

AUTH LOGIN

Supported web browsers:

Internet Explorer 9 and later, Mozilla Firefox 12 and later, Google Chrome 18 and later, Opera 11 and later

Recommended minimum screen resolution:

1024 x 768

Memory:

1000 values for each channel inside non-backup RAM memory

EMC emission:

EN 55022, Class B

EMC resistance:

EN 61000-4-2, levels 4/8kV, Class A

EN 61000-4-3, intensity of electromagnetic field 3V/m, Class A

EN 61000-4-4, levels 1/0.5kV, Class A

EN 61000-4-6, intensity of electromagnetic field 3V/m, Class A

Measured values

| Type | Temperature | Relative humidity | Pressure | Computed quantity |
|-----------|-------------|-------------------|----------|-------------------|
| EPD-T-INT | ✓ | | | |

| | | | | |
|--------------|---|---|---|---|
| EPD-TRH-INT | ✓ | ✓ | | ✓ |
| EPD-TRHP-INT | ✓ | ✓ | ✓ | ✓ |
| EPD-P-INT | | | ✓ | |
| EPD-TRH(P) | ✓ | ✓ | | ✓ |
| EPD-T | ✓ | | | |
| EPD-TRHP | ✓ | ✓ | ✓ | ✓ |

Temperature measurement

Temperature and humidity measuring range is limited - see graph below.

| Type | Operating temperature of electronic | Accuracy of | Range | Resolution |
|--------------|-------------------------------------|--------------------|------------------------------------|------------------|
| EPD-T-INT | -30 to +80°C
(-22 to +176°F) | ±0.6°C
(±1.1°F) | -30 to +80°C
(-22 to +176°F) | 0.1°C
(0.2°F) |
| EPD-TRH-INT | | | | |
| EPD-TRHP-INT | | | | |
| EPD-T | | ±0.2°C
(±0.4°F) | -200 to +600°C
(-225 to +999°F) | 0.1°C
(0.2°F) |
| EPD-TRH(P) | | ±0.4°C
(±0.7°F) | -30 to +105°C
(-22 to +221°F) | 0.1°C
(0.2°F) |
| EPD-TRHP | | | | |

Temperature response time with stainless steel sensor cover (F5200) in air flow 1m/s:

EPD-TRH-INT, EPD-TRH, EPD-TRHP-INT, EPD-TRHP
t₉₀ < 6min, temperature step 20°C (36°F)

EPD-TRHP t₉₀ < 16min, temperature step 20°C (36°F)

Temperature probe for Web Sensor EPD-T:

Pt1000/3850ppm, connected by shielded cable max. length 10m

Relative humidity measurement

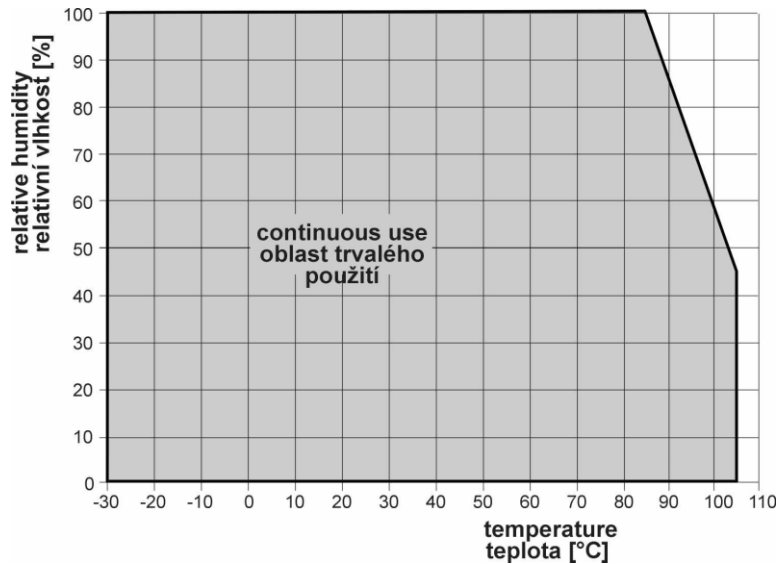
Temperature and humidity measuring range is limited - see graph below.

| Type | Operating humidity of electronic | Accuracy | Range | Resolution |
|--------------|--|---|---|------------|
| EPD-TRH-INT | 0 to 100%RH
without
condensation | ±2.5%RH
in range 5 to
95%RH
in 23°C (73.4°C) | 0 to 100%RH
temperature
compensated | 0.1%RH |
| EPD-TRHP-INT | | | | |
| EPD-TRH(P) | | | | |
| EPD-TRHP | | | | |

Humidity response time with stainless steel sensor cover (F5200) in air flow 1m/s:

t₉₀ < 30s (humidity step 65%RH, constant temperature)

Relative humidity and temperature restrictions:



Atmospheric pressure measurement

| Unit | Range | Accuracy | | |
|---------------------|-----------------|----------|--------------|-------|
| | | T = 23°C | 0 ≤ T ≤ 40°C | else |
| hPa | 600 to 1100 | ±1.3 | ±1.5 | ±2.0 |
| mBar | 600 to 1100 | ±1.3 | ±1.5 | ±2.0 |
| PSI | 8.70 to 15.95 | ±0.02 | ±0.02 | ±0.03 |
| mmHg | 450.0 to 825.1 | ±1.0 | ±1.1 | ±1.5 |
| inHg | 17.72 to 32.48 | ±0.04 | ±0.04 | ±0.06 |
| inH ₂ O | 240.9 to 441.6 | ±0.5 | ±0.6 | ±0.8 |
| ±oz/in ² | 139.2 to 255.3 | ±0.3 | ±0.3 | ±0.5 |
| kPa | 60.00 to 110.00 | ±0.13 | ±0.15 | ±0.20 |

Atmospheric pressure response time:

t90 < 44s (Response time is selectable. More detail you find in calibration manual, which is included with TSensor software installation.)

Computed quantities

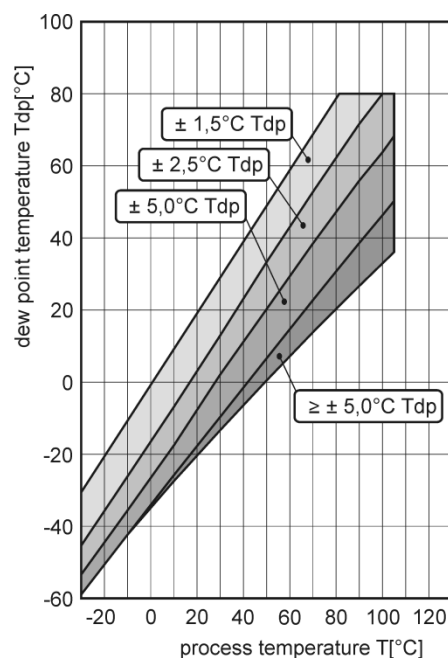
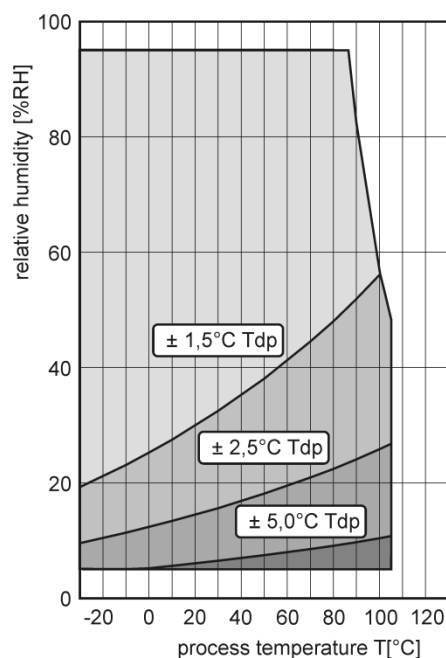
Dew point temperature

Range:

-60 to +80°C (-20 to 176°F)

Accuracy:

±1.5°C (±2.7°F) at ambient temperature T < 25°C (77°F) and RH > 30%, for more details see graphs:



Specific humidity

This value depends on atmospheric pressure. Pressure for quantities calculation is stored in device memory. Default value is 1013hPa and can be changed by software.

Range:

0 to 550g/kg

Accuracy:

$\pm 2.1\text{g/kg}$ at ambient temperature $T < 35^{\circ}\text{C}$ (95°F)

Mixing ratio

This value depends on atmospheric pressure. Pressure for quantities calculation is stored in device memory. Default value is 1013hPa and can be changed by software.

Range:

0 to 995g/kg

Accuracy:

$\pm 2.2\text{g/kg}$ at ambient temperature $T < 35^{\circ}\text{C}$ (95°F)

Specific enthalpy

This value depends on atmospheric pressure. Pressure for quantities calculation is stored in device memory. Default value is 1013hPa and can be changed by software.

Range:

0 to 995kJ/kg

Accuracy:

$\pm 4\text{kJ/kg}$ at ambient temperature $T < 25^{\circ}\text{C}$ (77°F)

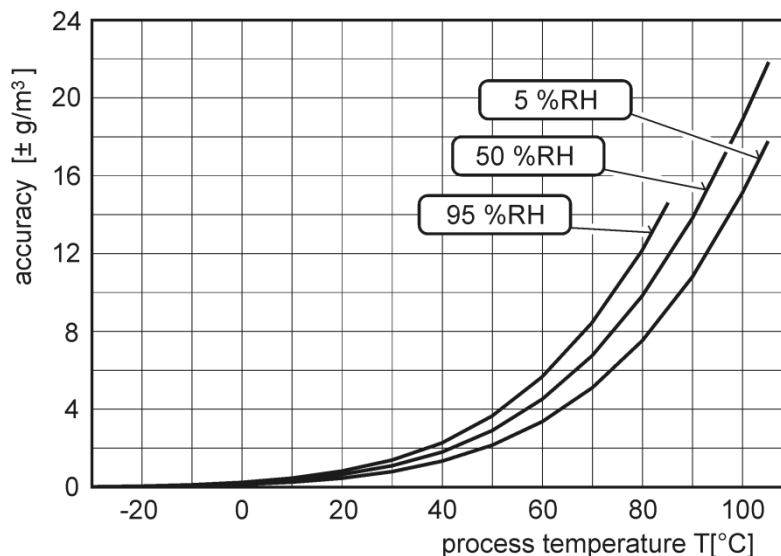
Absolute humidity

Range:

0 to 400g/m³

Accuracy:

$\pm 3\text{g/m}^3$ at ambient temperature $T < 40^\circ\text{C}$ (104°F), for more details see graph:



Operating terms

Temperature and humidity range in case with electronic:

-30°C to $+80^\circ\text{C}$, 0 to 100%RH (no condensation)

It is recommended to switch off in temperature over $+70^\circ\text{C}$.

Temperature and humidity range of external probe, cable including (EPD-TRH(P), EPD-TRHP):

-30 to $+105^\circ\text{C}$, 0 to 100%RH (no condensation, limited by graph)

Temperature of Pt1000 probe (EPD-T):

depends on probe model

Protection:

IP30 case with electronic

IP40 temperature-humidity probe (EPD-TRH, EPD-TRHP, EPD-TRHP)

Operating pressure range of probe EPD-TRHP:

up to 25bar

Air flow velocity around EPD-TRHP probe:

up to 25m/s at pressure of 1bar (1m/s at a pressure of 25bar)

Other environmental conditions (external influences) according to HD 60364-5-51:

normal

Working position EPD-P-INT, EPD-TRH(P), EPD-T, EPD-TRHP:

arbitrary

Working position EPD-T-INT, EPD-TRH-INT, EPD-TRHP-INT:

with sensor cover downwards. When mounting in RACK 19" with universal holder ATS-AC20 (optional accessories) then sensor cover can be placed horizontally.

End of operation



Disconnect the device and dispose it according to current legislation for dealing with electronic equipment. Electronic devices must be professionally destroyed in accordance with EU Directive 2002/96/ES of 27th January 2003.

Technical support and service

Technical support and service is provided by distributor. Contact is included in warranty certificate.

Preventive maintenance

Make sure the cables are not damaged periodically. We recommend you periodic calibration for measurement accuracy validation. Recommended calibration interval:

| Type | Calibration interval |
|--------------|----------------------|
| EPD-T-INT | 2 years |
| EPD-TRH-INT | 1 year |
| EPD-TRHP-INT | 1 year |
| EPD-P-INT | 1 year |
| EPD-TRH(P) | 1 year |
| EPD-T | 2 years |
| EPD-TRHP | 1 year |

Optional accessories

This chapter contains list of optional accessories, which can be ordered by extra cost. Manufacturer recommends using only original accessories.

Temperature probe Pt1000 for Web Sensor EPD-T

Wide ranges of Pt1000/3850ppm probes are available as optional accessories. Please contact your local distributor to help you with choice of correct probe according your application.

Power supply Adapter EPD-AC01

Power supply adapter with CEE 7 plug, 100-240V 50-60Hz/12V DC, 0.5A for Web Sensor devices.

Device case holder for RACK 19" ATS-AC20

ATS-AC20 is a universal holder for mounting of Web Sensor devices into RACK 19".

Probes holder for RACK 19" ATS-AC21

Universal holder for easy mounting probes in RACK 19".

Flow chamber SH-PP for Web Sensor EPD-TRHP

Flow chamber for compressed air measurement up to 25 bars. Made from stainless steel DIN 1.4301, inlet and outlet connection - G1/8, thread for humidity probe connection - G1/2.